

Active Battery Repositioning for Autonomous Center-of-Gravity Trim Optimization in Small UAVs with Variable Payloads

AME441 Project Proposal

Group 16

Fion Gawad | Alia Harris | Zuyi Li | Jack Nuttall

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Abstract

Small unmanned aerial vehicles (UAVs) commonly rely on elevator deflection to maintain longitudinal trim when payload changes or assembly variations shift the aircraft center of gravity (CG). Sustained elevator deflection can increase trim drag and actuator demand, potentially reducing flight efficiency and available control authority at low cruise speeds. Prior moving-mass research has demonstrated that longitudinal displacement of an internal mass can generate pitching moments and alter aircraft trim. This study investigates whether the existing battery of a small fixed-wing electric UAV can serve as a movable ballast mass that autonomously repositions the aircraft CG and reduces the elevator deflection required for longitudinal trim under variable and droppable payload conditions. A lead-screw-driven battery carriage will be commanded within pre-determined static-stability limits, while an IMU-based elevator controller maintains a prescribed pitch attitude. Wind-tunnel experiments will compare a conventional fixed-battery configuration with active battery repositioning under matched payload-induced CG disturbances. Elevator trim angle, battery displacement, pitch response, settling time, and drag will be measured. The resulting data will quantify the relationship between CG correction and elevator trim demand and evaluate whether active battery repositioning can reduce trim drag while preserving longitudinal stability.

I. Purpose

The primary objective of this project is to determine whether actively repositioning the battery as movable ballast in a small electric UAV can reduce steady-state elevator trim demand after changes in payload mass or location. The study will evaluate applications involving modular or droppable payloads. By shifting the CG toward an experimentally identified trim-optimal location while preserving a required positive static margin, the aircraft may maintain the commanded pitch attitude with less sustained elevator deflection, lower actuator demand, and potentially lower trim drag.

II. Scope

The team will design and build a simple fixed-wing UAV model with a rectangular high wing and a servo-actuated elevator on a 3D-printed horizontal stabilizer. A lightweight carriage mounted on linear rails inside the fuselage will carry the aircraft battery. Another carriage will translate the hinged pivot separate from the battery carriage. Two stepper motors and lead screws will translate the carriages longitudinally. The control electronics will include a microcontroller, IMU, stepper-motor driver, elevator servo, battery-position sensing through a limit switch and/or encoder, and a LiPo battery.

An opening in the lower fuselage will provide access to the carriage and mounting mechanism. The wind-tunnel support will connect to a hinged pitch pivot associated with a translating carriage. Before aerodynamic testing, the team will verify the relationship between pivot position, battery position, and calculated overall-aircraft CG so that gravitational moment about the support does not influence the aerodynamic trim measurement.

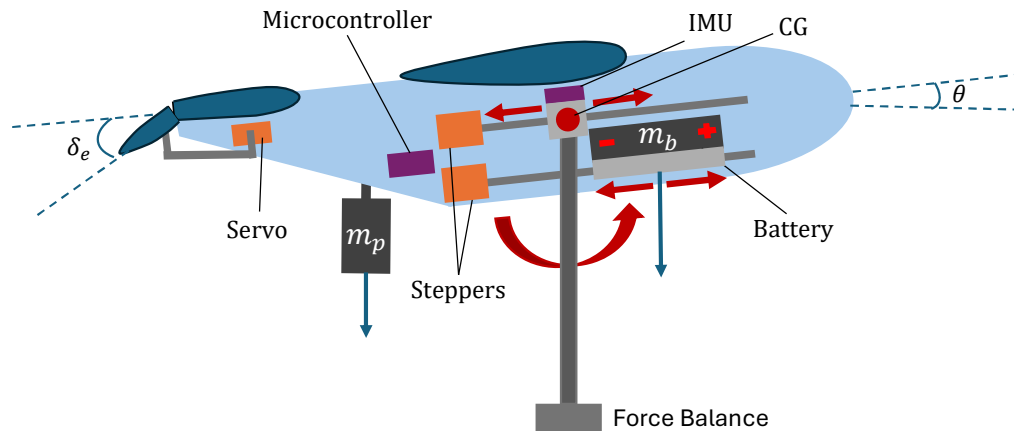


Figure 1. Experimental arrangement showing aircraft pitch angle ϑ , elevator deflection δe , payload disturbance mass m_p , and battery ballast mass m_b .

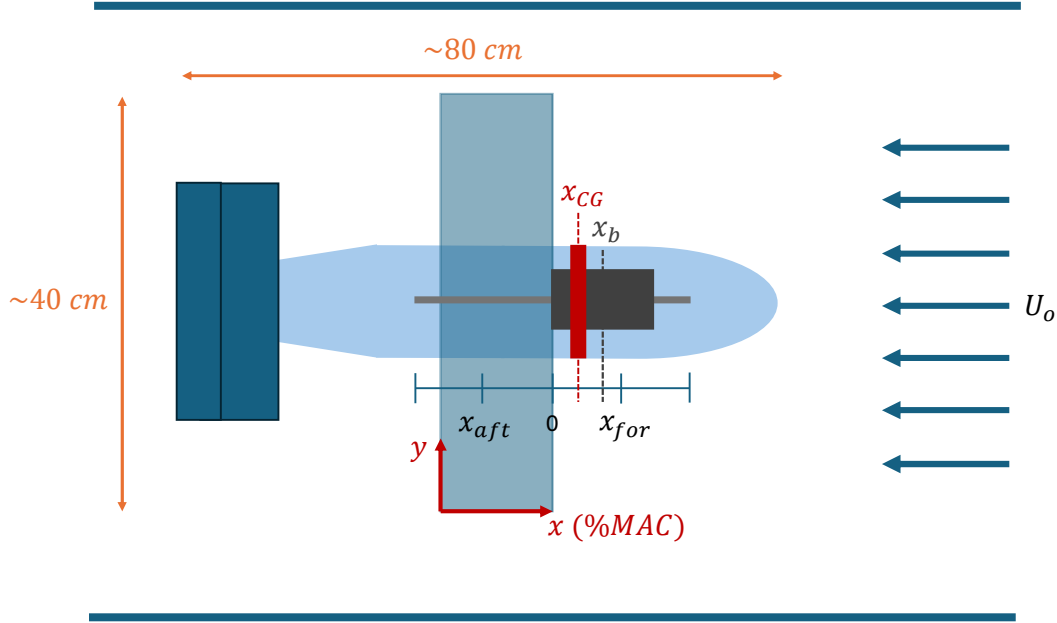


Figure 2. Wind-tunnel arrangement with estimated aircraft sizing, longitudinal battery-carriage coordinate system, location of CG x_{CG} , and location of battery x_b .

Performance will be evaluated by comparing the fixed-battery and active-repositioning configurations under identical air speed, payload, and commanded-pitch conditions. The primary response variables are steady-state elevator deflection and drag. Pitch error and pitch-rate variation will confirm that the compared cases achieve equivalent trim quality.

III. Technical Basis and Governing Equations

3.1 Center-of-Gravity Model

Following the moving-mass formulation of Ahmed et al. (2025), the center-of-mass position relative to a fixed body-frame datum is

$$\rho = \frac{1}{M}(m_f r_f + m_s r_s) \quad (1)$$

where M is total system mass; m_f and r_f describe the fixed portion of the vehicle; and m_s and r_s describe the movable mass. Applied along the longitudinal axis and extended to include the fixed airframe, movable battery, and payload, the aircraft CG is

$$x_{CG,0} = \frac{m_f x_f + m_b x_b + m_p x_p}{m_f + m_b + m_p} \quad (2)$$

The subscripts f , b , and p denote the fixed airframe, battery, and payload. All longitudinal positions will be measured from a common datum at the leading edge of the wing mean aerodynamic chord, with positive x directed aft.

3.2 Nondimensional CG and Static Margin

Following Zhai et al. (2023), longitudinal CG location will also be expressed as percent mean aerodynamic chord (MAC):

$$\bar{x}_{CG,0} = 100 * \frac{x_{CG,0} - x_{LE}}{\bar{c}} \quad (3)$$

where $\bar{c}$ is the mean aerodynamic chord and x_{LE} is its leading-edge location. Longitudinal static margin is

$$SM = \frac{(x_{NP} - x_{CG})}{\bar{c}} \quad (4)$$

where x_{NP} is the aircraft neutral-point location. The battery command will be constrained so that the CG remains within the predetermined forward limit and aft limit, $x_{CG} \leq x_{NP} - SM_{min} \bar{c}$.

3.3 Payload-Induced CG Shift and Required Battery Travel

Let $\bar{x}_{CG,0}$ denote the no-payload trim-optimal CG established during the initial sweep. After a payload is installed while the battery remains at its baseline position, the payload-induced CG shift is

$$\Delta x_{CG,p} = x_{CG,p} - x_{CG,0} \quad (5)$$

If the battery then moves by Δx_b while total mass remains constant, the resulting CG is

$$x_{CG,new} = x_{CG,p} + \frac{m_b}{M_0+m_p} \Delta x_b \quad (6)$$

where M_0 is the baseline aircraft mass including the battery. Setting $x_{CG,new} = x_{CG,0}$ gives the required battery travel:

$$\Delta x_b = \frac{M_0+m_p}{m_b} (x_{CG,0} - x_{CG,p}) \quad (7)$$

For a single added payload, this reduces to

$$\Delta x_b = -\frac{m_p}{m_b} (x_p - x_{CG,0}) \quad (8)$$

The negative sign indicates that the battery moves opposite the payload-induced mass moment. Predicted battery travel from Eqs. (7) and (8) will be compared with the empirically observed carriage position.

3.4 Trim Control and Measured Response

The IMU-based elevator controller will maintain a prescribed pitch attitude. Pitch error defined as

$$e_\theta(t) = \theta_{target} - \theta(t) \quad (9)$$

The commanded elevator deflection will be generated by a proportional-integral-derivative (PID) controller with pitch-rate feedback:

$$\delta_{e,cmd} = \delta_{e,bias} + K_p e_\theta + K_1 \int e_\theta dt - K_D q \quad (10)$$

where q is measured pitch rate from the IMU. A condition will be accepted as steady trim only after pitch error and pitch rate remain within specified tolerances for a to-be defined settling interval. At each stable condition, the mean elevator deflection and drag will be calculated. Because reduced elevator deflection is not itself a direct drag measurement, aerodynamic-efficiency conclusions will be based on force-balance drag data rather than elevator angle alone.

IV. Data and Experimental Method

Testing will begin with an initial CG sweep to identify a safe trim-optimal baseline. The elevator controller will maintain the same target pitch while the battery is fixed at seven discrete locations spanning the CG envelope. The aircraft will not be tested outside the prescribed static-stability range. The CG producing the minimum magnitude of steady elevator deflection will be designated $\bar{x}_{CG,0}$.

A second test block will add payloads at predetermined longitudinal stations while the battery remains fixed at its baseline location. Candidate payload configurations will target CG shifts of approximately $\pm 3\%$, $\pm 6\%$, and $\pm 9\%$ MAC relative to $\bar{x}_{CG,0}$. The final levels will be reduced if necessary to remain within the safe CG envelope and available elevator control authority. The tunnel will operate at one controlled freestream velocity selected during taring within the 10-20 m/s range.

The final steady-state block will repeat the same physical payload configurations with active battery repositioning enabled. The elevator loop will continue to maintain target pitch while the slower battery loop reduces the mean steady-state elevator command. Elevator deflection, drag, battery location, calculated CG, pitch error, pitch rate, and settling time will be recorded. Theoretical battery displacement from Eq. (8) will be compared with measured displacement.

V. Research Question

Can active repositioning of a UAV's existing battery autonomously compensate for variable payload-induced longitudinal CG shifts and reduce the elevator deflection and drag required to maintain stable trimmed flight?

VI. Facilities and Equipment

Testing will be conducted in the AME BHE Wind Tunnel, which has a $46 \times 46 \times 91$ cm test section and has been characterized at airspeeds from 10 to 30 m/s. The UAV model and mounting system will fit within these dimensions and have constrained pitch motion and battery travel. The force balance will support the model

and measure drag. Testing will begin at low speed and progress to a single controlled test velocity between 10 and 20 m/s after sufficient elevator control authority and structural behavior are verified.

The experimental system will include the UAV model, movable battery carriage, linear rails, lead screw, stepper motors and driver, elevator servo, microcontroller, IMU, LiPo battery, payload masses, and custom wind-tunnel mount. Required components and estimated costs are summarized in Figure 3.

AME-441 Group 16 - Cost Estimate Active Battery Repositioning for Autonomous Center-of-Gravity Optimization in Small UAVs as of 9/3/2026					
	Vendor/Manufacturer	Model/Part No.	Qty.	Unit Price	Item Price
<u>EXPENSES FOR ITEMS AVAILABLE IN LAB</u>					
Manufacturing					
Laser Cutting (\$50 per hour)	AME-441 Machine Shop	DWG-GRP-X-03R2	0.5	\$50.00	\$25.00
3D Printing with PLA (\$1.00 per cubic inch)	AME-441 Machine Shop		100	\$1.00	\$100.00
Facilities					
Wind Tunnel Rental (\$450 per hour)	AME-441 Lab		10	\$450.00	\$4,500.00
				Sub-total Cost:	\$4,625.00
<u>EXPENSES FOR ITEMS REQUIRED FOR PURCHASE</u>					
Equipment					
Micro Stepper Motor	STEPPERONLINE	8HS15-0604S	2	\$25.08	\$50.16
Stepper Motor Driver	BIGTREETECH	TMC2209 V1.3	1	\$22.09	\$22.09
Arduino UNO R4 Minima	Arduino	ABX00080	1	\$19.99	\$19.99
3S LiPo Battery	Zeee	11.1V 50C 2200mAh	1	\$34.19	\$34.19
XT60 Pigtales	GINTOOYUN	B0CBKH3Q7H	1	\$15.99	\$15.99
Lever Wire Connectors	Shutao	2in8	1	\$6.99	\$6.99
Gear Servo Motor	Hitec RCD	HRC32065S	1	\$35.99	\$35.99
UBEC Servo Voltage Regulator	ZORZA	CY358	1	\$9.99	\$9.99
9-DOF IMU	Adafruit	BNO085	1	\$29.50	\$29.50
Power Switch	DAIERTEK	KCD1-5-101	1	\$6.39	\$6.39
Materials					
Bearings	McMaster-Carr	60355K151	2	\$5.97	\$11.94
Linear Rails	uxcell	B0D54LGWHJ	1	\$27.29	\$27.29
Carbon Steel Shaft Rod	McMaster-Carr	1327K114	1	\$4.67	\$4.67
Lead Screw	YEJMKJ	T6x2-400x2	1	\$18.49	\$18.49
				Sub-total Cost:	\$293.67
				TOTAL PROJECT EXPENSES:	\$4,918.67
<u>BUDGET</u>					
AME-441 project budget (\$100 per student)			4	\$100.00	\$400.00
				TOTAL BUDGET:	\$400.00

Figure 3. Bill of materials and estimated project cost.

VII. Test Matrix

The test matrix uses matched payload cases to isolate the effect of active battery repositioning. Three repeated runs at each performance condition will support repeatability and provide uncertainty. Calibration and tare measurements will be completed before and after each test block as appropriate.

Table 1. Proposed wind tunnel test matrix for evaluating the aerodynamic and dynamic effects of active center-of-gravity (CG) repositioning.

Group	Diagnostic / Goal	Battery Mode	Conditions	Repeats	Total
0	Sensor calibration and tunnel tare	Fixed	Wind off; model-support tare at test speed	Before/after blocks	N/A
1	Map CG position to trim elevator and drag; select $x_{CG,0}$	Fixed at commanded locations	Seven safe CG positions at approximately 2-2.5% MAC spacing within the approved envelope	3 each	21
2	Establish no-payload baseline at selected trim-optimal CG	Fixed at $x_{CG,0}$	No payload; constant target pitch and airspeed	3	3
3	Measure uncompensated payload-induced CG disturbance	Fixed at baseline battery position	Six payload cases targeting $x_{CG} = \pm 3\%$, $\pm 6\%$, and $\pm 9\%$ MAC, subject to safety limits	3 each	18
4	Evaluate active compensation under matched disturbances	Active	Same six physical payload configurations as Group 3; battery allowed to reposition	3 each	18
	Total performance runs				60

VIII. Safety Concerns

Several safety concerns must be considered for the active center-of-gravity repositioning system. The primary concern is that moving the battery too far or too quickly could induce an unstable flight condition by

shifting the aircraft's center of gravity outside the stable range or creating a rapid pitching response. The aircraft's CG location and neutral point will therefore be determined through aerodynamic analysis and research before testing to establish a safe operating range for battery movement. The maximum battery travel and movement rate will be limited to keep the aircraft within this stable range. Additional concerns include the LiPo battery and electrical components, which will be securely mounted and inspected before testing to prevent damage or electrical failures. Emergency on-off switches will also be incorporated for the electrical system.

Since this experiment will allow the aircraft to pivot about the battery carriage, the test stand will also be designed to limit excessive pitch rotation. Before conducting full wind-tunnel tests, the system will be tested at low airspeeds to verify structural integrity, actuator operation, and safe battery movement.

IX. Project Schedule

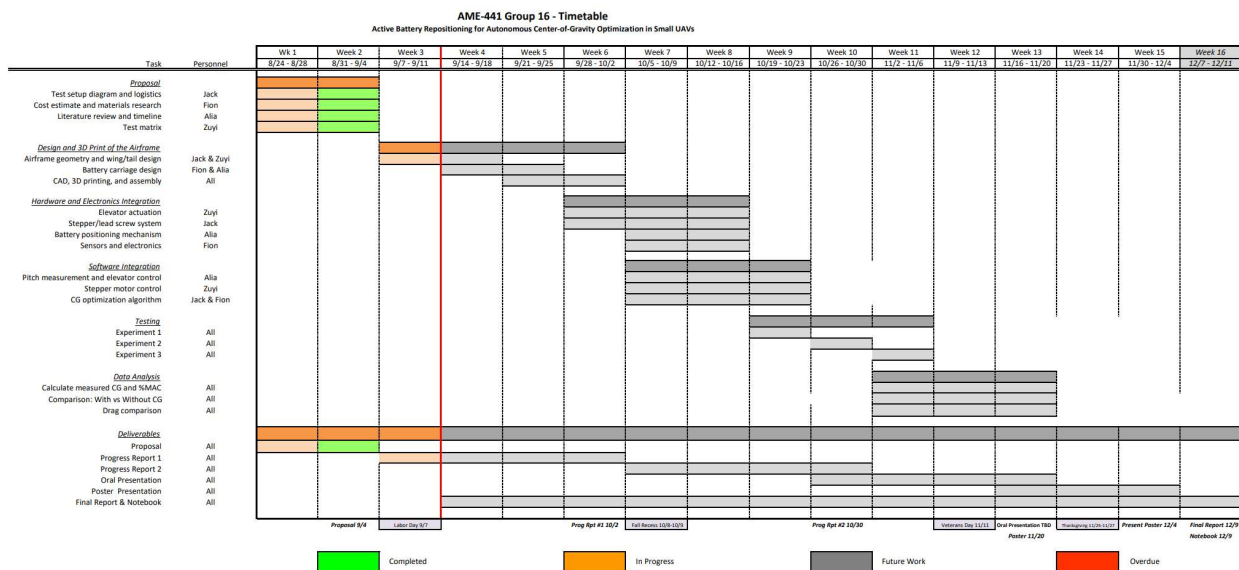


Figure 4. Projected timeline for project completion.

X. Sources

- Ahmed, Anas, et al. "Dynamic Center-of-Mass Reconfiguration for Enhancement of UAV Performances Based on Simulations and Experiment." *Drones*, vol. 9, no. 12, 12 Dec. 2025, article 854, <https://doi.org/10.3390/drones9120854>. Accessed 18 June 2026.
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